	REQUIREMENTS	TYPE OF TESTING(I = Integration Testing S = System Functionality Testing O = Operational Acceptance Testing UN = Unit Testing US = Usability Testing A = Acceptance Testing,- = not applied)	PASS OR FAIL	CONTRIBUTOR	STATUS
	USER REQUIREMENTS				
	INSTRUCTOR				
1.1	Must log in securely and access administrative functions	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,BAO	☑
1.2	Must be able to create, update, enable, disable, and manage instructor accounts	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,BAO	☑
1.3	Must be able to revoke instructor access when required.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,BAO	☑
1.4	Must be able to monitor system health and operational status.	UN,I,S,US,A,O	PASS	BAO	☑
1.5	Must be able to review audit logs and security-related events.	UN,I,S,US,A,O	PASS	BAO	☑
1.6	Must be able to review failed Canvas and AI operations.	UN,I,S,US,A,O	PASS	RHYTHM	☑
1.7	Must be able to troubleshoot the system without accessing sensitive credentials.	UN,I,S,US,A,O	PASS	BAO	☑
	ADMINISTRATOR				
2.1	Must be able to securely create account, authenticate and access the application	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
2.2	Must be able to securely connect a Canvas API token.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,BAO	☑
2.3	Must be able to replace or remove a stored Canvas API token.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,BAO	☑
2.4	Must be able to view Canvas courses associated with the connected token.	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,BAO	☑
2.5	Must be able to select an active course.	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,BAO	☑
2.6	Must be able to view course information, students, assignments, and submissions.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,BAO,RHYTHM	☑
2.7	Must be able to analyze missing submissions.	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,BAO	☑
2.8	Must be able to analyze late submissions.	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,BAO	☑
2.9	Must be able to search and filter course information	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,BAO	☑
2.10	Must be able to access a dashboard summarizing course activity.	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,BAO	☑
2.11	Must be able to interact with the system using natural-language requests.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
2.12	Must be able to ask questions about course data.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
2.13	Must be able to receive explanations of proposed actions before execution.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
2.14	Must be able to create Canvas assignments.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
2.15	Must be able to update assignment details and dates.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
2.16	Must be able to create Canvas quizzes.	UN,I,S,US,A,O	PASS	JESSICA	☑
2.17	Must be able to list submissions for assignments	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,BAO	☑
2.18	Must be able to draft Canvas messages to selected students.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
2.19	Must be able to review and edit messages before sending.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
2.20	Must be able to draft feedback comments for selected submissions	UN,I,S,US,A,O	PASS	JESSICA	☑
2.21	Must be able to review and approve feedback before posting.	UN,I,S,US,A,O	PASS	JESSICA	☑
2.22	Must be able to modify grades through approved workflows.	UN,I,S,US,A,O	PASS	MRIDUL	☑

2.23	Must be able to preview Canvas-modifying actions before execution.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
2.24	Must explicitly approve Canvas-modifying actions before execution.	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,YERDANA	☑
2.25	Must be able to cancel a proposed action before execution.	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,YERDANA	☑
2.26	Must be able to view a history of previous actions.	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,YERDANA	☑
2.27	Must be able to view previous assistant requests and their outcomes.	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,YERDANA	☑
2.28	Must be able to determine whether a previous action can be reversed	UN,I,S,US,A,O	PASS	MRIDUL	☑
2.29	Must receive clear feedback when requests cannot be completed safely.	UN,I,S,US,A,O	PASS	RHYTHM,MRIDUL,YERDANA	☑
	FUNCTIONAL REQUIREMENTS				
	Authentication, Account Management, and Canvas API Token Setup				
1.1	The system must support instructor login before any Canvas-related functionality is accessible.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,BAO	☑
1.2	The system will allow administrators to create, update, enable, disable, and manage instructor accounts.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,BAO	☑
1.3	The system will allow administrators to revoke instructor access.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,BAO	☑
1.4	The system will allow instructors to enter, replace, or remove a Canvas API token.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,BAO	☑
1.5	Canvas API tokens must be encrypted before storage.	UN,I,S,US,A,O	PASS	MRIDUL,BAO	☑
1.6	The system must validate Canvas API tokens before use.	UN,I,S,US,A,O	PASS	MRIDUL,BAO	☑
1.7	Stored Canvas API tokens must never be exposed to users after submission.	UN,I,S,US,A,O	PASS	MRIDUL,BAO	☑
1.8	Canvas operations must be performed using the authenticated instructor's stored Canvas API token.	UN,I,S,US,A,O	PASS	MRIDUL,BAO	☑
1.9	The system must prevent one instructor from accessing another instructor's Canvas API token.	UN,I,S,US,A,O	PASS	MRIDUL,BAO	☑
1.10	The system must support logout and session expiration.	UN,I,S,US,A,O	PASS	MRIDUL	☑
	Course Dashboard and Canvas Data Access				
2.1	The system will retrieve courses available to the connected instructor.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
2.2	The system will allow instructors to select an active course.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
2.3	The system will retrieve students for the selected course.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,RHYTHM,BAO	☑
2.4	The system will retrieve assignments for the selected course.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
2.5	The system will retrieve submissions for selected assignments.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
2.6	The dashboard will display course-level summary information.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,RHYTHM,BAO	☑
2.7	The dashboard will display assignment-level information and submission status.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
2.8	The system will identify submitted, missing, late, and graded submissions where Canvas provides the required information.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
2.9	The dashboard will provide suggested actions and workflow shortcuts.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
2.10	Canvas will be treated as the source of truth for course and assignment information.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
	Natural-Language Assistant				
3.1	The system will provide a chat-style natural-language interface.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
3.2	The assistant will support approved Canvas course-management workflows.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
3.3	The assistant will interpret user requests and convert them into structured actions.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
3.4	The assistant will request clarification when information is ambiguous or incomplete.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
3.5	The assistant will display the interpreted intent before execution.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑

3.6	The assistant will support: o Listing courses o Listing students o Listing assignments o Listing submissions o Creating assignments o Updating assignment dates o Drafting student messages o Drafting feedback comments o Creating quizzes o Modifying grades	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
3.7	The assistant will explain unsupported requests and prevent execution.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
3.8	The assistant must not invent Canvas information that has not been retrieved from Canvas.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
3.9	The assistant will scope requests to the selected active course unless another course is explicitly specified.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
	Assignment and Quiz Management				
4.1	The system will support creation of Canvas assignments.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
4.2	The system will support updating assignment details including assignment dates.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
4.3	The system will support creation of Canvas quizzes.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
4.4	Assignment and quiz information must be validated before submission to Canvas.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
4.5	Assignment and quiz creation or modification must require preview and confirmation.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
	Submission Analysis and Grading				
5.1	The system will identify students with missing submissions for a selected assignment.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
5.2	The system will identify students with late submissions for a selected assignment.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
5.3	The system will display submission summaries and affected student lists.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO	☑
5.4	The system will support grade modification workflows through approved Canvas operations.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL	☑
5.5	Grade modifications must require preview and confirmation.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL	☑
	Canvas Messaging and Feedback				
6.1	The system will support generation of draft Canvas messages.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
6.2	Instructors will be able to review and edit messages before sending.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
6.3	Message delivery must require preview and confirmation.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
6.4	The system will support generation of text feedback comments for selected submissions.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
6.5	Feedback posting must require preview and confirmation.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
6.6	Messaging and feedback actions must be recorded in action history.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
	Dry-Run Preview and Confirmation				
7.1	Every Canvas-modifying action must generate a dry-run preview before execution.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
7.2	The preview must summarize the proposed action and affected Canvas resources.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
7.3	The preview must identify whether an action is reversible.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
7.4	Users must explicitly confirm Canvas-modifying actions before execution.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
7.5	Users must be able to cancel proposed actions.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑
7.6	The system will classify actions as read-only, modifying, or irreversible.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,YERDANA	☑

	Action History and Limited Reversal				
8.1	The system must maintain a user-visible action history.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO,YERDANA	☑
8.2	Action history must include the original request, action performed, execution	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO,YERDANA	☑
8.3	status, timestamp, and affected Canvas resources.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO,YERDANA	☑
8.4	The system will maintain a history of assistant requests and associated actions.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,RHYTHM,BAO,YERDANA	☑
8.5	The system will support limited reversal of supported actions where safe and practical.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO,YERDANA	☑
8.6	The system must explain when reversal is unavailable.	UN,I,S,US,A,O	PASS	MRIDUL,RHYTHM,BAO,YERDANA	☑
	AI Validation and MCP Integration				
9.1	The AI model must not directly access Canvas APIs.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,RHYTHM	☑
9.2	All Canvas operations must be routed through the backend and MCP/tool-server layer.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,RHYTHM	☑
9.3	The backend must validate AI-generated actions before execution.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,RHYTHM	☑
9.4	The backend must reject malformed, unsupported, ambiguous, unauthorized, or unsafe actions.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,RHYTHM	☑
9.5	The MCP/tool-server layer will expose only approved Canvas operations.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,RHYTHM	☑
9.6	The project must include a Dockerized MCP server or equivalent tool-execution layer.	UN,I,S,US,A,O	PASS	JESSICA,MRIDUL,RHYTHM	☑
	Security and Privacy				
10.1	Canvas API tokens and AI API keys must be encrypted at rest.	UN,I,S,US,A,O	PASS	JESSICA,RHYTHM,BAO	☑
10.2	Sensitive credentials must never appear in logs, error messages, API responses,repositories, or user interfaces.	UN,I,S,US,A,O	PASS	JESSICA,RHYTHM,BAO	☑
10.3	Student information must only be visible to authorized users.	UN,I,S,US,A,O	PASS	JESSICA,RHYTHM,BAO	☑
10.4	The system must minimize student-identifiable information sent to AI services.	UN,I,S,US,A,O	PASS	JESSICA,RHYTHM,BAO	☑
10.5	The backend must validate all Canvas-modifying actions before forwarding requests to Canvas.	UN,I,S,US,A,O	PASS	JESSICA,RHYTHM,BAO	☑
	Reliability and Error Handling				
11.1	The system must provide meaningful user-facing error messages.	UN,I,S,US,A,O	PASS	JESSICA,RHYTHM	☑
11.2	The system must safely handle Canvas API failures.	UN,I,S,US,A,O	PASS	JESSICA,RHYTHM	☑
11.3	The system must safely handle incomplete or unavailable Canvas data.	UN,I,S,US,A,O	PASS	JESSICA,RHYTHM	☑
11.4	The system must prevent duplicate execution of Canvas-modifying actions.	UN,I,S,US,A,O	PASS	JESSICA,RHYTHM	☑
	Administration and Deployment				
12.1	The system will provide administrators with operational health information.	UN,I,S,US,A,O	PASS	BAO	☑
12.2	The system will support monitoring of failed Canvas and AI operations.	UN,I,S,US,A,O	PASS	BAO	☑
12.3	The application will be implemented as a web-based system.	UN,I,S,US,A,O	PASS	BAO	☑
12.4	The project will include a Docker Compose configuration for local deployment.	UN,I,S,US,A,O	PASS	JESSICA	☑
12.5	The project will include developer setup and deployment documentation.	UN,I,S,US,A,O	PASS	JESSICA	☑
12.6	The system architecture will separate the frontend, backend, database, Canvas integration layer, and MCP/tool-server components.	UN,I,S,US,A,O	PASS	JESSICA	☑
	Scope Constraints				
13.1	The MVP will not support autonomous Canvas modifications without explicit instructor approval.	S,A	PASS	JESSICA,MRIDUL,RHYTHM,BAO,YERDANA	☑
13.2	The MVP will not support direct Canvas API execution that bypasses backend validation.	S,A	PASS	JESSICA,MRIDUL,RHYTHM,BAO,YERDANA	☑
13.3	The MVP will not support institution-wide Canvas administration.	S,A	PASS	JESSICA,MRIDUL,RHYTHM,BAO,YERDANA	☑

13.4	The MVP will not support unrestricted deletion of Canvas resources.	S,A	PASS	JESSICA,MRIDUL,RHYTHM,BAO,YERDANA	☑
13.5	The MVP will not support arbitrary user-defined Canvas API calls.	S,A	PASS	JESSICA,MRIDUL,RHYTHM,BAO,YERDANA	☑